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Academic Sandbox: Enabling Access and Exploration Mobile Positioning Data for Supporting Mobility Analytics

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***Abstract*—With the advancement of technology and the increasing use of mobile devices, large amounts of data are generated as a byproduct. One such dataset is Mobile Positioning Data (MPD), derived from Call Detail Records (CDR) collected by mobile network providers, which may estimate both individual and collective human mobility. MPD holds valuable insights into movement patterns and offers potential for solving various interdisciplinary challenges. However, its sensitive nature and proprietary restrictions limit access for academic researchers. To address this challenge, we propose the development of an academic sandbox system. This system leverages Kibana for interactive visualization and Elasticsearch for data storage and analysis, enabling researchers to explore MPD and derive insights without compromising raw data confidentiality. The system offers three dashboards to meet diverse research needs while protecting proprietary data: an overview dashboard displaying daily displacement, visit frequency, and travel distance; an origin-destination (OD) dashboard with OD matrix, arc diagrams, and tables; and an urban hotspot dashboard showing individual movement within districts or regions. This work creates new opportunities for research related to MPD-based mobility studies, such as investigating the impact of human movement patterns on the spread of COVID-19.**

***Index Terms*—people mobility, mobile positioning data, mobility visualization, data analytics, academic sandbox, data accessibility, data exploration**

I. Introduction

As one of the most crucial electronic devices in modern human life, mobile phone usage has become very prevalent. Globally, consumers spent 5.1 trillion hours interacting with it [1]. These mobile interactions can have various purposes, from entertainment to communication, and will produce various data as the byproducts during the process.

Call Detail Record (CDR) data is one of the byproducts gathered by telecommunication companies when a mobile device or a Mobile Station (MS) interacts with their telecommunication network [2]. In general, CDR contains users' information (e.g. user identification (UID)) and transaction details (e.g. connection time, transaction type, or relevant cell tower's area code) which are then used as parameters to calculate the bills for each user who uses the telecommunication company's services.

Using geolocation processes, like triangulation and trilateration [3], companies can further extract estimated location of their users from the CDR they have gathered and produce a set of data called Mobile Positioning Data (MPD). Although users' locations contained in the resulting MPD is only an estimate, MPD still has a lot of usages not only for the telecommunication companies but also for other parties, like scientists and researchers, as well.

One of the use cases where MPD can be valuable for scientists and researchers is for people mobility analysis. This subject is mainly dedicated to observe and study human's mobility and series of movements that create a certain trajectory pattern, whether on individual or collective level. While individual trajectories can provide us with specific information about a person, e.g. their home and work address [4] or their mode of transport [5], it can only be analysed statistically as its stochastic nature makes it hard to be predicted accurately [6]. Only when we have a lot of individual trajectories data and look at it as a collective can we do a more accurate mobility analysis prediction, e.g. representative group clustering based on different activity types [7].

In certain study and real world cases, e.g. global pandemic handling, trip route optimization, or density estimation, people mobility pattern and the relevant MPD data become an integral part to the solutions. However, although they hold an essential role, user location data like MPD is still part of the telecommunication companies' confidential data and tied to their users' privacies, meaning not everyone can have access to it. Processing a huge amount of MPD data also requires a lot of resources which cannot be afforded by most individual scientists and researchers, especially when they're still in the exploration stage. A sandbox platform can be the solution to this issue.

Sandbox is a scalable development platform which can be used by third-party users to explore aggregate data and information without touching the confidential raw data in

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an interactive and collaborative way [8]. This environment provides the computational resources needed for data scientists to perform complex analysis which can be hard to do if done conventionally. Data sandbox allows researchers to see trends and extract insight from Big Data as quickly as possible.

The contribution of this paper is the proposal of academic sandbox designed to facilitate the exploration of citizen movement through the use of Mobile Positioning Data (MPD) The research also focuses on data engineering techniques to manage and process large-scale MPD Data. The rest of the paper is organized as follows. In Section 2, we present related works on mobility analysis as well as a review of the current methodologies. In Section 3, we discuss research methodology. Results and discussions are provided in Section 4. Finally, we conclude our work in Section 5.

II. Related Works

A. Mobility Analysis

Human mobility analysis can be achieved by using either one of two data sources: Call Detail Records (CDR) and Global Positioning System (GPS). CDR produces estimated user location data (MPD) by utilizing transaction details and additional information generated during cell phone communication with the telecommunication network, while GPS relies on the hardware system which has its tracking feature enabled when a variety of smartphone apps or other devices is running [9].

Although its accuracy highly depends on the telecommunication network's cell tower density and there's a certain concern for users' privacy, MPD produced using CDR has a wider spatial coverage compared to GPS [10]. On the other hand, GPS produces high-resolution geolocation data which is more precise albeit having narrower spatial coverage and possibility of user bias [9].

Multiple people mobility analysis research based on MPD has been conducted prior, e.g [11]. Although based only on number of visits and average travel distance, the analyses done in the aforementioned research still needed the access to the relevant MPD data. Therefore, this research highlights the importance of an interactive dashboard to allow scientists and researchers to gather insights without directly accessing the data itself due to its sensitivity.

B. Preprocessing Procedures

Preprocessing procedures for geotemporal (location-time) data is not a new topic and has been a focus of studies in multiple fields, e.g. in transportation modelling [12], trajectory data analysis [13], and to assist with minimizing location uncertainty during smart navigation development [14].

1) Preprocessing in Transportation Model: In this transportation model research, the paper focuses on proposing a set of experimentation procedure as a method to decide the most appropriate zoning system and immobility threshold as the required initial assumptions and parameters before complex mobility analysis, e.g. origin-destination (OD) mapping [12]. The data source used was an already cleaned MPD data with no nulls or redundancy, minimized error, and normalized values to represent the actual population within the region. Therefore, the method in the aforementioned research cannot suffice as the only preprocessing steps used for this case where the MPD data is still raw.

2) Preprocessing in Trajectory Data Mining: The preprocessing method proposed in this trajectory data mining and analysis paper consists of 5 important steps: noise filtering, stay point detection, trajectory compression, trajectory segmentation, and map matching [13]. For each step, this paper also offers a few possible algorithms which can be used to achieve the desired outcome, e.g. mean filter algorithm for noise filtering and Douglas-Peucker for trajectory compression. The paper also demonstrates how the 5-steps preprocessing procedure can minimize data inaccuracy and produces data which is as close to reality as possible. Despite this, the emphasize on high accuracy makes the process offered cannot be used as the sole reference in this case where the analysis would focus on collective mobility instead of individual trajectory.

3) Preprocessing to Minimize Local Uncertainty: To minimize local uncertainty of their geotemporal data, Xu *et al.* proposed a preprocessing method based on Mean Absolute Deviation (MAD) which is claimed to show significant improvement compared to the more common methods such as two-stage clustering and time window-based which are most effective used to identify representative location and to minimize location oscillation respectively. In this research, the data used is called Mobile Signaling Data (MSD) which records users' locations on cell tower levels continuously, even when they are inactive and not interacting with their mobile phone. This differs slightly from MPD which is based on CDR data generated only during mobile interactions, but similar enough that applying this method to MPD should also be possible albeit the effectiveness is unknown. With that said, the preprocessing method proposed is only focused on minimizing uncertainty and the result is not yet ready for analysis, making it unsuitable to be the only reference used for the preprocessing method in this case.

III. Methodology

This section outlines the research design, data collection methods, and analytical techniques employed in this study.

A. Platform/Tools

We employ the complete ELK stack, consisting of Elasticsearch, Logstash, and Kibana, due to its adaptability, allowing for customization and feature extension to meet specific requirements. Additionally, as an open-source solution, it provides access to a large, active community of users and developers, facilitating collaboration and continuous improvement. Elasticsearch is used to index and search through MPD, perform complex queries, and extract valuable insights. Logstash is used to ingest events to Elasticsearch, ensuring that they are clean, standardized, and ready for analysis. Finally Kibana is used to provide visualizations, enabling researchers to gain meaningful insights from MPD.

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B. Data source

The mobile positing data is derived from positioning systems of mobile device collected by network providers. We have an access to the location data of hashed cell phone collected by one of the biggest network providers in Indonesia. The data was collected from March to June 2020. This time frame coincided with the implementation of Large-Scale Social Restrictions (PSBB) or lockdown measures, which could potentially impact human mobility.

C. Data preprocessing

The raw MPD data obtained from Elasticsearch consists of $16,493,947$ instances. Several preprocessing steps were performed on this data.

1) Cleaning. Data cleaning involves addressing duplication and null values. This process includes removing inconsistencies, errors, and outliers within the dataset. Handling null values is particularly important to address missing or incomplete data points and ensure the integrity of the analysis.
2) Thresholds determination: The `ImmobilityThreshold` represents the minimum duration an individual must remain within a zone to be considered in a state of immobility or to have reached their destination. In this paper, the `ImmobilityThreshold` is defined as 1 hour and 3 hours. The `deltaMAD` parameter is used to eliminate oscillation effects, with a value of `deltaMAD` set to 0.5. To filter out abnormal movements, a maximum speed threshold of $120km/h$ (`maxSpeed`) is applied.
3) Stay detection. To detect the `stay state` from several nearby points indicating immobility, a two-stage clustering approach is utilized:
   a) Outlier Removal: This stage involves eliminating consecutive identical latitude and longitude readings and filtering out abnormal movements exceeding the `maxSpeed` threshold.
   b) Clustering: The process begins by selecting an initial point in the trajectory, defined by its latitude and longitude coordinates ($lati$, $longi$), and an initial distance (so). The algorithm iterates through each point in the dataset, calculating the distance between the current point and the initial point. If the difference between this distance and the initial distance exceeds a specified threshold, `roamingDistance` $1.5km$, the current point becomes the new center of a new cluster, updating $lati$, $longi$, and so accordingly. If the distance difference is within the threshold, the point is added to the current cluster. This approach effectively groups trajectory points into clusters based on their spatial proximity, creating clusters that represent areas of close proximity within the trajectory data.
4) Oscillation removal.
   Oscillation patterns refer to the detected movement of a location due to the alternating signal detection from multiple cell towers (BTS) even when the device remains stationary. The MAD (Mean Absolute Deviation) method is employed to eliminate oscillation patterns, specifically oscillation pattern 1 (A-B-A) and oscillation pattern 2 (A-B-B-A), which result in the frequent appearance of a particular location (A) within an individual's trajectory. The underlying idea of this approach is that each individual typically visits a limited number of locations within a day, so a realistic trajectory should have a MAD within a certain range. Trajectories affected by oscillation tend to have a significantly higher MAD than this range, as the related cells are recorded repeatedly in the Mobile Signalling Data (MSD). Thus, the MAD approach aims to iteratively remove these oscillation sequences until the MAD value converges.
   We employ the Mean Absolute Deviation (MAD) as outlined in Equation 1,
   $$MAD = \frac{\sum_{i=1}^{n} |x_i - m(X)|}{n} \tag{1}$$
   where n represents the total number of locations in an individual's daily trajectory, x_i denotes the frequency of visits to the ith location, and $m(X)$ refers to the median frequency across all locations.
   The clustering results and oscillation removal process yielded $6,069,656$ instances that are now ready for further processing.
5) Reverse geocoding is performed to identify administrative areas based on latitude and longitude coordinates. This process utilizes GeoPandas[1] to accurately map geographic coordinates to their corresponding administrative boundaries. The resulting dataset enables further analysis within the specified administrative contexts.

D. Metrics

In order to support mobility analytics, we calculate several metrics as follows.

1) Travel Distance and Num. of Visit: Travel Distance is defined as the distance between two latitude longitude coordinates using an inverse computation approach, in this work, we use Pyproj[2] library. The number of visits is calculated based on the day and time by determining the frequency of movements and the average distance of each movement. The results of calcuation will be shown in `Overview Dashboard`.

2) Origin-Destination: To obtain a clearer understanding of the movement between districts, we identify the origin points and destination points for each district.

Furthermore, to provide a more comprehensive overview of inter-district movement in matrix form, additional processing is performed on the MPD origin-destination data. This involves deriving matrices for the originating districts and destination districts The resulting data frames are then reintroduced into Elasticsearch with the index 'mpd_od_overview'. The results

[1]https://geopandas.org/en/stable/
[2]https://pyproj4.github.io/pyproj/stable/api/geod.html

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of the Origin-Destination matrix will be presented in the `OD Dashboard`.

IV. Results and Discussion

A. Visualization

Academic sandbox provides three level dashboards. Each dashboard visualizes Mobile Positiong Data in different aggregation levels. Therefore, each dashboard can be used in different use cases.

1) Overview Dashboard: informs the total number of displacement during a period. It also contains three bar charts that may inform people about mobility patterns, i.e., (the total number of daily displacement per day and its regression line, median of individual daily displacement, dan individual average travel distance) as shown in Figure 1. This dashboard may be used by researchers interested in daily movement in particular time. User may change the observation time period on the dashboard.

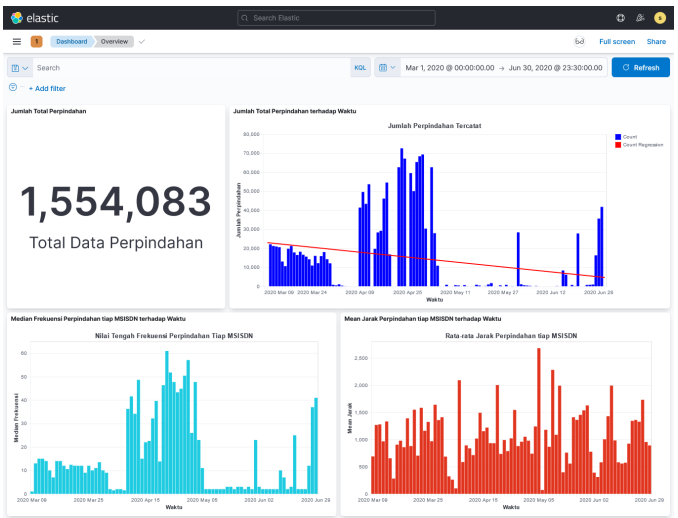

Fig. 1. Overview Dashboard

2) Origin-destination (OD) dashboard: visualize mobility patterns in the level of regencies and districts. OD dashboard contains three information, i.e., an OD matrix between regencies, an OD arc between districts, and a table showing total number of daily displacement between districts as in Figure 2.

3) Urban hotspot dashboard: provides detailed information on the population density of a district in the Yogyakarta Province through a choropleth or shaded area as shown in Figure 3. This dashboard shows further down to the individual level by displaying the MSISDN of people present in that area during a specific time period.

B. Security

The academic sandbox has distinct aggregation and visualization details on each dashboard. This distinction requires varying levels of authorization for access. We implement access control based on roles (Role-based access control).

TABLE I
Query time in displaying graphs during March-June 2020

Dashboard	Visualization	Avg. Query Time (ms)
Overview	Total number of daily displacement and linear regression	602.8
	Median visit frequency	727.8
	Mean travel distance	668.1
OD	OD Matrix	400.8
	OD Arc	807
	OD Table	431.4
Urban hotspot	Urban hotspot choropleth	2981
	MSISDN table	523.1

The overview and OD dashboards are accessible for users with general 'researchers' role. The Urban hotspot dashboard contains more granular information, such as number of people in a district or regency. Further, in hotspot dashboard, a user may obtain information of the number of daily displacement specific per user in a district/regency, represented as MSISDN (Mobile Subscriber Integrated Services Digital Network Number). While the MSISDN has been replaced by a value to ensure the actual MSISDN is not exposed, the urban hotspot dashboard is more restricted to limited role, such as 'senior researchers'.

C. Response time

The quality performance of the academic sandbox in this project is measured by the response time taken by Elasticsearch to provide a query response and by Kibana to visualize that response according to the selected graph type. In total, 10 queries will be executed for each experiment, and the results will be averaged to obtain the most accurate test results possible as shown in Table I.

D. Insight from dashboard

When a user utilizes Academic Sandbox, they may access dashboards depending on their role. They can modify visualization parameters. In this study, two customizable parameters are time and location. These modifications can be made to the OD dashboard and the urban hotspot dashboard using the time picker in Kibana. Figure 4 shows the bar chart of the total number of displacements per day. It can be seen that in March 2020, the trend of movement was decreasing, reaching only about 10,000 movements per day. This pattern was followed by very limited travel during the first week of April 2020. In April 2020, there was a significant increase in displacements until the end of the month, reaching 70,000 movements per day. This may be due to the decision of the Yogyakarta government not to implement large-scale social restrictions (PSBB) during this period [15]. People mobility then disappeared in May and June 2020. The reason might be the central government's release of the policy on the implementation of "Learning from Home" through Circular Letter number 15 of 2020 in May 2020 [16]. The trend of the number of displacement during the observation period is downtrend, as shown by red line (regression line) in Figure 4.

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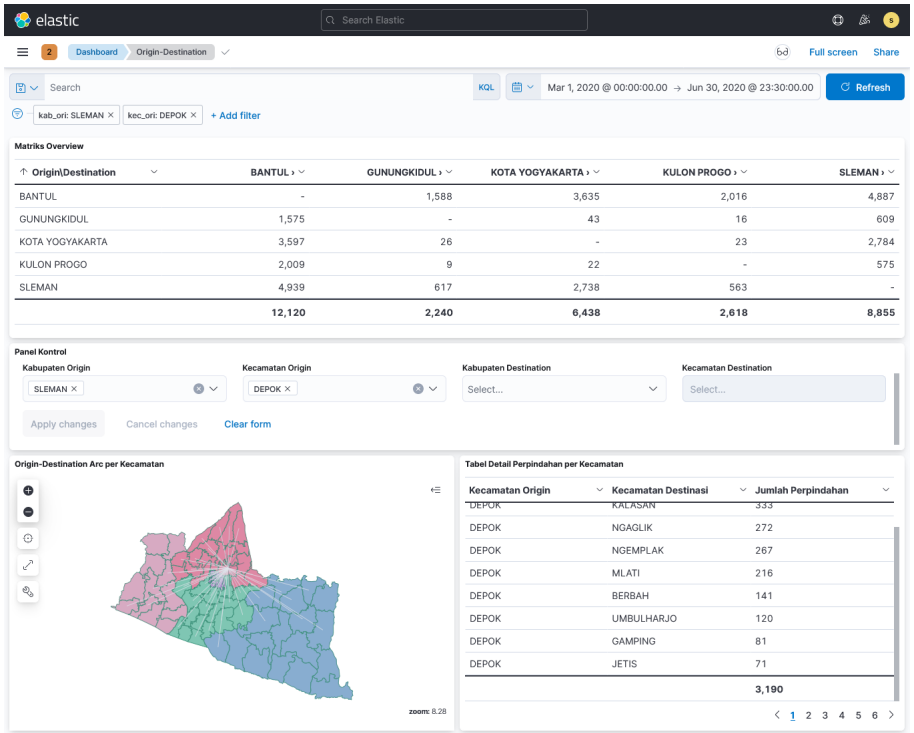

Fig. 2. Origin-Destination Dashboard: Designed to visualize movement flow

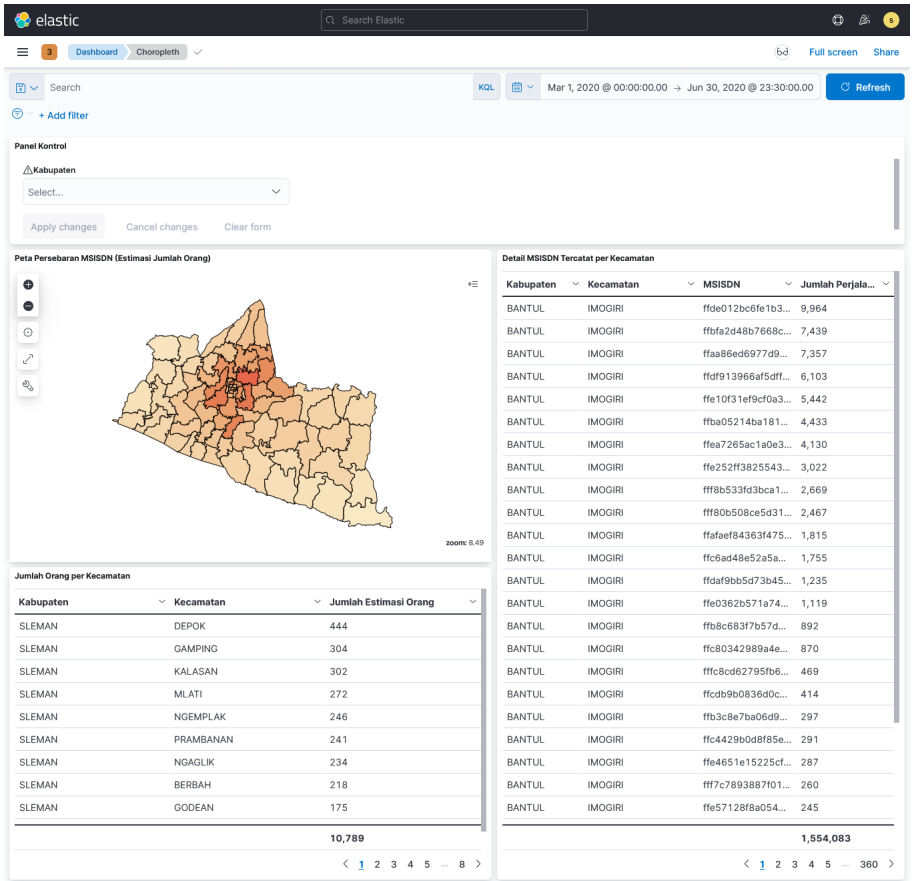

Fig. 3. Urban Hotspot Dashboard: Designed to visualize population density

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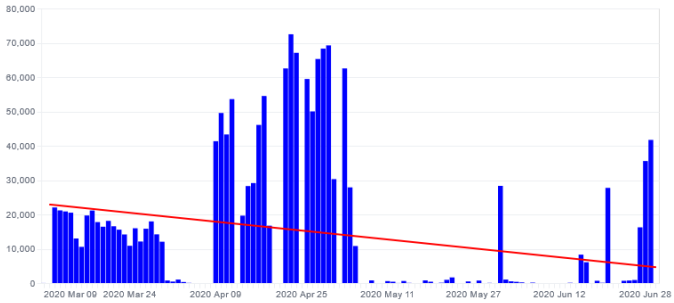

Fig. 4. Overview Dashboard: Displays the total number of displacements

V. Conclusion

The development of the Academic Sandbox as a tool for exploring Mobile Positioning Data (MPD) has proven to be highly effective for academic research. The tool offers three levels of dashboard functionality accessible depending on the users' role: an overview for general mobility statistics, an origin-destination flow analysis, and an urban hotspot visualization that illustrates population density. These features enable more researchers to gain comprehensive insights into mobility patterns and trends while ensuring the preservation of individual privacy.

To facilitate the analysis, a preprocessing pipeline was implemented to transform raw transactional data into analytic-ready data. This preprocessing ensures that the data can be efficiently processed and visualized within the sandbox environment, with front-end performance in Kibana demonstrating acceptable response times ranging from 300 to 807 milliseconds, depending on the type of visualization requested. This efficiency is crucial for enabling real-time data exploration and decision-making.

The customizable query feature in the Academic Sandbox enhances its usefulness by allowing users to adjust their analyses based on time and location. This flexibility was demonstrated in the analysis of daily displacement trends during the early COVID-19 pandemic in Yogyakarta. The study found a general downtrend in displacements, highlighting the impact of regional and national policies on population movement during this period.

Future development could focus on creating a custom web frontend to improve expericence. This is due to limitation on customizing Kibana's user interface. Additionally, we have noted the limited variety of data exploration options currently offered by the sandbox. Further studies and surveys on the use cases of MPD data could be conducted, and the findings could then be implemented to enhance and expand the system.

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